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Predation-mediated edge effects reduce survival of wader nests at a wet grassland-forest edge

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Abstract

Edge effects occur when the matrix has adverse impacts on the patches of remnant habitat. A widely explored example of this is the hypothesis of a higher predation pressure on bird nests closer to the habitat edge. In parallel with the recent loss of open habitats through afforestation as a climate change mitigation measure, an interest in the impact of forest on species dependent on open habitats has re-emerged. We follow wader nest survival to study the issue of an edge effect in a system of wet grasslands fragmented by forests in a region where it has not been tested before, focussing mainly on northern lapwing (*Vanellus vanellus*), common ringed plover (*Charadrius hiaticula*), common redshank (*Tringa totanus*) and southern dunlin (*Calidris alpina schinzii*). To record nest survival, we monitored 753 nests of 10 wader species on coastal grasslands in Estonia for 3 consecutive years. A subset (n = 85) of these nests was equipped with camera traps to record nest predation events and predator association with forest edge. The distance to nearest trees and forest and a forest cover within a 1-km buffer around each nest was measured. We recorded extremely low daily nest survival rates (0.903–0.922 for different species), with most nests lost to predation. We showed that nest survival is lower closer to the forest edge and negatively affected by the proportion of forest within a 1-km buffer around each nest. Based on camera trap recordings, we suggest that the edge effect is caused by elevated nest predation rates by the most common predator, the red fox (*Vulpes vulpes*), closer to the forest edge. Future afforestation plans of open habitats need to acknowledge that the resulting fragmentation has a negative impact on nest survival of ground-breeding birds. On the other hand, our results imply that restoration efforts aimed at removal of most damaging forest plantations could benefit breeding waders.

Introduction

Current processes of habitat conversion and climate change are depleting biodiversity on a large scale (Díaz *et al.,* 2019). Open landscapes are one of the habitats that have been altering at an unprecedented rate over the past century (Wauchope *et al.,* 2017; Terskaia *et al.,* 2020). Warmer temperatures and reduced management of semi-natural landscapes are causing regrowth (Burnside *et al.,* 2007; Rannap *et al.,* 2017) and shrubification of this habitat (Bignal & McCracken, 1996; Tape *et al.,* 2006). Large afforestation campaigns, as a climate change mitigation measure, reduce the extent of open habitats (Burrascano *et al.,* 2016) and growing anthropogenic pressures continue to increase the fragmentation of these already threatened areas (Sutherland *et al.,* 2012). As a result of these processes, waders, a group of threatened birds dependent mostly on open habitats, are losing their breeding and non-breeding grounds (Lindström & Agrell, 1999; Pearce-Higgins *et al.,* 2017; Gunnarsson, 2020). In addition to the direct risks posed by habitat loss, the resulting fragmentation contributes to the increase of habitat edge.

A wave of studies since the 1980s have illustrated the adverse effect of an edge: if the habitat edge is used by predators living in the surrounding biotopes, the edge may have a negative effect on the survival of the prey closer to the edge (Angelstam, 1986; Andrén & Angelstam, 1988; Paton, 1994; Renfrew & Ribic, 2003). One such example is the widely discussed and tested hypothesis that avian nest survival increases with the distance to habitat edge (Paton, 1994; Andrén, 1995; Chalfoun *et al.,* 2002; Batáry & Báldi, 2004; Stephens *et al.,* 2004). The knowledge on lower nest survival in habitat edge relies mostly on research carried out in forests (Andrén, 1995; Batáry & Báldi, 2004), while

similar studies from open landscapes have often not supported the edge effect theory (Andrén & Angelstam, 1988; Berg, 1996; Bertholdt *et al.,* 2017; Ottvall *et al.,* 2005; Vickery *et al.,* 1992; Valkama & Currie, 1999; but see: Ellis *et al.,* 2020; Johnson & Stanley, 1990; Møller, 1989). Nevertheless, there is strong evidence that species that breed in open habitats avoid forest (Stroud *et al.,* 1990; Hancock *et al.,* 2009; Wilson *et al.,* 2014; Kaasiku *et al.,* 2019) and may experience enhanced population declines in more fragmented landscapes (Valkama & Currie, 1999; Amar *et al.,* 2011; Douglas *et al.,* 2014; Franks *et al.,* 2017).

Birds breeding in open landscapes nest mainly on the ground, and such species have been shown more likely to be in decline, possibly because they are more vulnerable to predation (McMahon *et al.,* 2020). Therefore, the current processes that increase edge habitat may be especially damaging for ground nesting species, such as waders. As habitat fragmentation is a global issue, there is an increasing need also for understanding the geographical scope and scale of edge effects, as existing studies have mostly been carried out in a small range of regions.

Boreal Baltic coastal grasslands are semi-natural primeval wet pastures that have persisted for centuries through the combined effect of floods, winter ice, postglacial land uplift and grazing by domestic livestock and wild geese (Jutila, 2001; Auffret & Cousins, 2018). Among other benefits, these grasslands serve as an important wader breeding habitat. In the 1960s, Estonian coastal grasslands were described as 'endless plains reaching the horizon' characterized by the abundance of northern lapwing (*Vanellus vanellus*), southern dunlin (*Calidris alpina schinzii)* and common redshank (*Tringa totanus*), with black-tailed godwit (*Limosa limosa*), Eurasian curlew (*Numenius arquata*), common snipe (*Gallinago gallinago*) and ruff (*Calidris pugnax*) also mentioned as common breeding birds (Kumari, 1966). During the second half of the 20th century, land abandonment and an active commercial forestry campaign to afforest open habitats of 'low value', effectively damaged Estonian coastal grasslands (Palang, 1998; Luhamaa *et al.,* 2001). Between the 1960s and 2000s the area of coastal grasslands shrunk by *c* 70% (Luhamaa *et al.,* 2001) and now the remaining sites merge into forests in the higher parts of the coastal plains. At present, the Boreal Baltic coastal grassland habitat is listed in Annex I of the EU Habitats Directive as a priority habitat type (code 1630* European Commission, 2013). Despite efforts to restore this habitat by reducing reed and tree cover over the last two decades and the agri-environment scheme (AES) applied to support their management, most wader populations dependent on this habitat have not recovered (Estonian Environmental Agency, 2019). This indicates that the threatened wader populations may be limited by factors other than habitat availability, such as low productivity through increased predation (Roos *et al.,* 2018), which could possibly be exacerbated by the increase of habitat edge in these fragmented landscapes (Møller, 1989; Johnson & Stanley, 1990; Ellis *et al.,* 2020).

By studying a diverse breeding wader community, we aim to (i) fill the knowledge gap on edge effects in a wet grassland-forest system at edge and surrounding landscape scale in Baltic coastal grasslands and (ii) determine nest predators and their association with forest edge to understand the possible mechanism behind the edge effect.

Materials and methods

Study area

We recorded nest survival in 10 Boreal Baltic coastal grassland sites in Estonia (Fig. 1). These coastal grasslands are located in the hemi-boreal climate zone, characterized by cold winters (mean temperature in December–February −2.6 C°), warm summers (mean temperature in June–August 16.4 C°) and moderate precipitation (mean 662 mm/year; Estonian Weather Service, 2015). The study area included some of the most important and largest coastal grassland complexes in the country, as well as rather isolated and smaller sites, and also intermediate areas. The surveyed area per site ranged from 19 ha to 91 ha (mean = 41 ha, total = 408 ha). All study sites were managed under an AES by livestock grazing (mainly cattle, only one site had a mixed herd of horses and cattle). The sites varied in their microhabitats (differences in sward heterogeneity and height) but were all in accordance with the AES requirements which state that 50% of the area needs to be managed with short swards (up to 10 cm). Wader breeding densities at the study sites vary from 7 to 160 (mean = 61; calculated based on territory mapping in 2018 and 2019) wader pairs per km^2 – while some of the sites can be considered high-density breeding areas at the European level, the densities are low elsewhere (Kaasiku, unpublished). The typical wader species breeding in the coastal grassland habitat are the northern lapwing, common ringed plover (*Charadrius hiaticula*), common redshank, southern dunlin, common snipe, ruff, black-tailed godwit and Eurasian curlew (Estonian Environmental Agency, 2019).

Wader nest survival

We monitored wader nest survival over three breeding seasons (2018–2020). As the onset of the breeding season depends on breeding phenology of each species and its end is related to environmental conditions and breeding success, the length of the three seasons varied (2018: 21 April–8 July; 2019: 16 April–28 June; 2020; 02 April–03 July). The beginning of the survey was marked by the estimated clutch initiation by the species with the earliest phenology, the northern lapwing and the end of the season was considered when there were no known nests left in the area.

We searched for nests of all breeding wader species mostly by (i) scanning the landscape with binoculars and/or scope for incubating individuals and (ii) walking through the site to flush birds and visually detect nests with eggs. The fieldwork was mostly conducted by two people (ranging from 1 to 4). Upon finding a nest, the species was identified, two eggs (if available) were floated to determine the incubation stage according to Liebezeit *et al.* (2007), and

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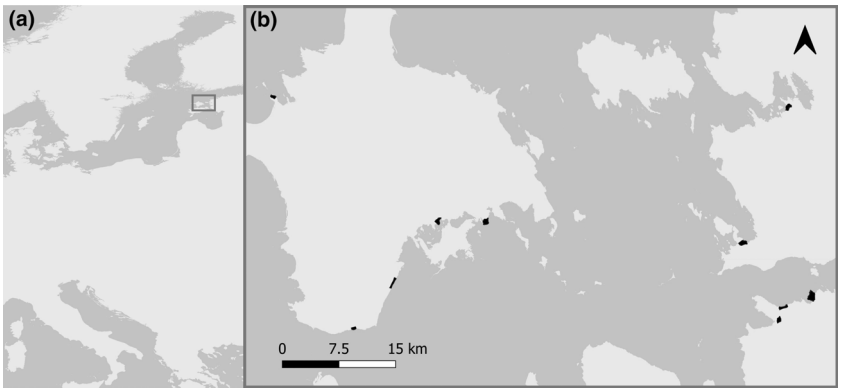

Figure 1 (a) Study location in Europe; (b) 10 study sites marked as black polygons.

nest location was recorded with OruxMaps application (oruxmaps.com) on Android devices. Each site was revisited after an average of 8 days (range 5–11 days). During the revisit, nests were monitored, and the status (warm eggs, depredated or hatched) of the nests registered using the same devices.

The assessment of the nest fate at the time of data collection may differ slightly from study to study, but as the selected approach can have a strong impact on the results, we provide a thorough overview of our methodology. We considered the nest to be hatched if the predicted hatching date was within a week from the visit and there were visible signs of hatching in the nest cup: the nest contained very small eggshell fragments that were clearly distinguishable from depredated shells. The nest was registered as depredated if there were no signs of hatching, that is, there were no small egg fragments in the nests cup or/and there were depredated eggshells in the vicinity of the nest. The nest was classified as abandoned if the incubation stage had not progressed between two visits and the eggs were cold (eggs in the nest were only floated again if the predicted hatching date had passed or if the nest seemed abandoned, i.e. had an unorganised look, the eggs seemed dull). The nest was registered as flooded, if there were signs of recent floods and intact eggs were found in the vicinity of the nest. Similarly, the nest was classified as trampled if there were visible trampling signs (e.g. footprints) and egg remains. The nest was classified with an unknown fate if the cause of the nesting failure could not be determined. Such examples occurred when: (i) no intact eggs were found after a flooding event, so it was not possible to distinguish between flooding and predation and (ii) in case the nest cup was destroyed by trampling in such way that it was impossible to check it for eggshells, meaning we could not tell if the nest had been hatched, depredated or trampled. When failing to re-find the nest, the area was searched again during the next visit. If the nest could not be located during two consecutive visits, knowing that it should not have hatched within that time, it was considered as depredated for species with open nests and unknown for species with concealed nests, but if the nest should have hatched during this time, it was classified as with an unknown result independent of the nest type.

As predation was the major cause of the nest failure, and the aim of the study was to assess the effect of predation on nest survival, nests lost for other causes were excluded from further analysis.

Nest predation and predator detection rates

We registered predation events and predator detection rates by using observations recorded by camera traps set at 85 wader nests (brands ScoutGuard, OUVision, Bushnell, Browning) in 2019 and 2020. The mean camera distance to the forest edge was 334 m (min = 20, max = 870, 95% CI = 284–384 m). The cameras were used from the beginning of the nesting season until the release of cattle or horses (mostly in the end of May), because cattle tend to show interest in the cameras, increasing the risk of nest trampling and pushing the equipment over. Cameras were deployed when the nest was found and removed at the next visit after hatching or nest loss. Cameras were placed at one little ringed plover (*Charadrius dubius*), eight redshank, 12 common ringed plover and 64 lapwing nests. Cameras were not deployed at nests of species of higher conservation concern to avoid any risk of the devices affecting nest survival (e.g. southern dunlin, black-tailed godwit). Cameras were placed *c* 3 m from the nest and attached to a short wooden stick at the height of *c* 30 cm. Cameras were set to start recording 20 sec videos or take three images when the motion sensor was triggered (selected media type depended on the capacity of the memory card), following a 1-min break before allowed to be triggered again. Camera traps recorded predation events and predator detection rate in an index unit predator observation per camera day.

Landscape characteristics

We looked at the effect of habitat on nest survival at edge and landscape scale. Considering different scales can help explain the variation in nest predation (Chiavacci

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et al., 2018; Ellis *et al.,* 2020). For edge-scale analysis we mapped trees and forest in or adjacent to the study sites from aerial images taken in May 2018 (pixel size 20×20 cm; Estonian Land Board, 2019). A patch of trees or bushes was classified as 'tree' if the width of the patch was less than 30 m, all woodland patches wider than that were classified as 'forest'. Distances between each nest and the closest tree and forest edge were measured. To identify the effect of forest cover at a landscape scale, a buffer of 1 km around each nest was created and the percentage of the area under forest cover inside this buffer was calculated. For this, we used the extent of forest cover obtained from the base map of the Estonian Land Board (2020). The landscape scale approach should provide information on the habitat use by predators, hence we used the territory size of the most common mesopredator in the area, the red fox (*Vulpes vulpes*), as the landscape metric. The size of selected buffer indicates the size of the red fox territory in a similar wet grassland habitat in Denmark (314 ha; Meisner *et al.,* 2014). All spatial processes were carried out in QGIS 3.8.1.

Analyses

Nest survival analysis was carried out with MARK programme's RMark interface (Dinsmore & Dinsmore, 2007; Laake, 2011) and nest survival was expressed as daily survival rates (DSR). The following factors: distance to forest, distance to trees, forest cover (proportion of forest within a 1-km buffer around each nest), date of clutch initiation (back calculated based on the incubation stage; given in days from the 1st of March each year), year, species and the interaction between distance to forest and species, were included in the model as covariates (Table 1). We also included models that tested the effect of the year on distance to forest, distance to trees and forest cover. The resulting models were compared based on the AIC_c values. Overall DSR values were calculated by the null model.

To test the variation of forest cover and distance to trees and forest between the years, as well as to study nest predation events in relation to forest edge and forest cover by different species, we used the Kruskal–Wallis test. To test the effect of forest cover, distance to forest and year on predator detection rates, we used these variables in a generalized linear mixed model using template model builder (glmmTMB) in the lme4 package. Wilcoxon matched pairs test was used to compare red fox observations in active and inactive nests. All statistical analyses were performed using R (R Core Team, 2020).

Table 1 Descriptions of covariates used in the nest survival model

Covariate	Unit	Description
Forest	m	Distance to forest (range = 20–1021 m)
Tree	m	Distance to trees (range = 12–896 m)
ForestCover	%	Proportion of forest within a 1-km buffer around each nest (range = 0–0.5)
LayDay		Days from the 1st of March each year
Year		Year of the study
Species		Lapwing, c. ringed plover, redshank, dunlin, godwit, ruff, snipe, oystercatcher, curlew, l. ringed plover
Forest:Species		Interaction of distance to forest and species

Results

Nest survival

During three breeding seasons we monitored 753 nests of 10 wader species: northern lapwing (lapwing; n = 444), common ringed plover (C. ringed plover; n = 139), common redshank (redshank; n = 91), southern dunlin (dunlin; n = 54), black-tailed godwit (godwit; n = 8), ruff (n = 5), common snipe (snipe; n = 5), Eurasian oystercatcher (*Haematopus ostralegus;* oystercatcher; n = 5), Eurasian curlew (curlew; n = 1) and little ringed plover (L. ringed plover; n = 1). Two hundred and thirteen nests hatched, 526 failed, and in 14 the outcome was undetermined. Eighty-nine per cent of the nests that did not hatch were lost to predation, while other causes of nest failures included abandonment (6.7%), flooding (2.3%), trampling (0.2%) and an unknown reason (2.3%). The seemingly high nest abandonment rate may also include flooded nests in case the flooding event was not detected and can be resulted from the overall high predation rates (when partially predated and subsequently abandoned nests in an early incubation stage were included in the dataset as nests in the laying stage). Our final analysis was based on 679 nests that either hatched or were depredated.

The DSR over all site-years was 0.929 (SE = 0.0032, 95% CI = 0.922–0.935; separate values for most common species are given in Table 2), which with a combined incubation and laying duration of 27 days (average across our sample) would result in 14% (95% CI = 11–16%) probability of surviving incubation. The distance to forest (X^2 = 3.493, df = 2, P-value = 0.174) and trees (X^2 = 1.428, df = 2, P-value = 0.490) and forest cover (X^2 = 5.062, df = 2, P-value = 0.080) did not differ between years.

Top two MARK models explaining variance in nest survival differed <2 AIC_c units and included covariates (i) forest cover and year, and (ii) distance to forest and year (Table 3). These two models combined accounted for nearly 100% of the model weight (67% and 33%, respectively). If year is excluded as a covariate, distance to forest and forest cover again almost equally explain the variance in nest survival, although performing >10 AIC_c units worse than the models that include the year. DSR is higher further from the forest and where the forest cover is lower (Table 4; Fig. 3). To illustrate the values from Fig. 3 numerically, 7% (95% CI = 5–11%) of the nests located closest to the forest (at 20 m) hatch, while the hatching rate is 26% (18–34%) for the nests furthest from the forest (at 1020 m). Similarly, 3% (1–8%) of nests hatch when the forest cover is 0.5 and 19% (15–23%) hatch at lowest forest cover (0). DSR does not vary by species, by laying date, nor by the interaction of distance to forest and

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Table 2 DSR values (Estimate), standard error (SE), lower (LCL) and upper (UCL) confidence limits for four of the most common species

Parameter	*n*	Estimate	SE	LCL	UCL	Laying+incubation (days)	Probability of surviving incubation
Dunlin	51	0.932	0.0126	0.903	0.953	25	0.17
C. ringed plover	123	0.937	0.0068	0.922	0.949	27	0.17
Redshank	83	0.925	0.0095	0.904	0.942	27	0.12
Lapwing	398	0.929	0.0042	0.92	0.937	28	0.13

Estimated combined laying and incubation duration is given to calculate species-specific proportions of hatched nests (Probability of surviving incubation = Estimate $^{(\text{Laying + incubation})}$). Laying duration is estimated to be 1 egg/day, incubation starting when the fourth egg is laid. Incubation duration is derived from Soikkeli (1967), Wallander (2003), Glutz von Blotzheim et al. (1977, via Thyen & Exo, 2005) and own data (N = 8), for the species presented in the table, respectively.

Table 3 Daily survival models evaluating the effect of six covariates: distance to nearest forest (Forest), forest cover within a 1-km buffer around each nest (ForestCover), distance to nearest trees (Tree), Year, the onset of the laying in days from the 1st of March on each study year (LayDay), Species and the interaction of distance to forest and species (Forest:Species)

Model	Parameters	AIC_c	ΔAIC_c	Weight	Deviance
ForestCover + Year	**4**	**1457.89**	**0.00**	**0.63**	**1449.88**
Forest + Year	**4**	**1458.93**	**1.05**	**0.37**	**1450.93**
Tree + Year	4	1470.37	12.48	0.00	1462.36
Forest	2	1470.84	12.95	0.00	1466.84
ForestCover	2	1471.15	13.26	0.00	1467.15
Year	3	1471.66	13.77	0.00	1465.66
Tree	2	1480.84	22.96	0.00	1476.84
Forest:Species	17	1481.33	23.44	0.00	1447.20
Null	1	1481.91	24.02	0.00	1479.91
LayDay	2	1482.48	24.59	0.00	1478.48
Species	9	1483.99	26.10	0.00	1465.95

The models are ranked by their performance, with the two competing models that best explain the variation in daily survival rates presented in bold.

Table 4 Beta estimates, standard error (SE), lower confidence limit (LCL) and upper confidence limit (UCL) for top two nest survival models

Model	Parameter	Estimate	SE	LCL	UCL
ForestCover + Year	Intercept	2.9263	0.1203	2.6906	3.1621
	ForestCover	**−1.6587**	**0.4069**	**−2.4563**	**−0.8612**
	Year2019	0.0156	0.1331	−0.2453	0.2765
	Year2020	**−0.4081**	**0.1282**	**−0.6595**	**−0.1568**
Forest + Year	Intercept	2.3942	0.1274	2.1445	2.6439
	Forest	**0.0007**	**0.0002**	**0.0003**	**0.0011**
	Year2019	0.0434	0.1328	−0.2168	0.3037
	Year2020	**−0.3710**	**0.1271**	**−0.6201**	**−0.1218**

Values for other models are presented Table S1. Estimates for Year are in relation to 2018. Effects marked in bold have confidence intervals that do not overlap zero.

species (Table S1). As a limitation, these analyses do not account for the possible spatial autocorrelation in nest survival, which can have a density-dependent relationship (e.g. MacDonald & Bolton, 2008a, but see also Galbraith, 1988). However, as presented in Supplementary material (Figure S1), the high predation rates cause temporal variation in active nest timings in our dataset, making it difficult to account for spatial autocorrelation.

Nest predation and relative predator abundance

The red fox predation accounted for 48% of the depredated nests, golden jackal 8%, raven (*Corvus corax*) 5% and badger (*Meles meles*) 3%, while in 36% of the 64 predation events in total, the camera malfunctioned and failed to record the nest predator (Fig. 2). We do not have any reason to

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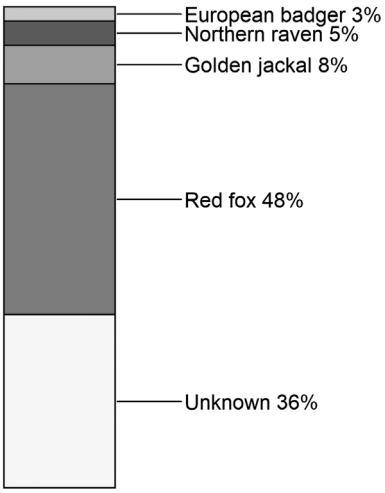

Figure 2 Nest predator proportions recorded by camera traps at 64 nests that were depredated.

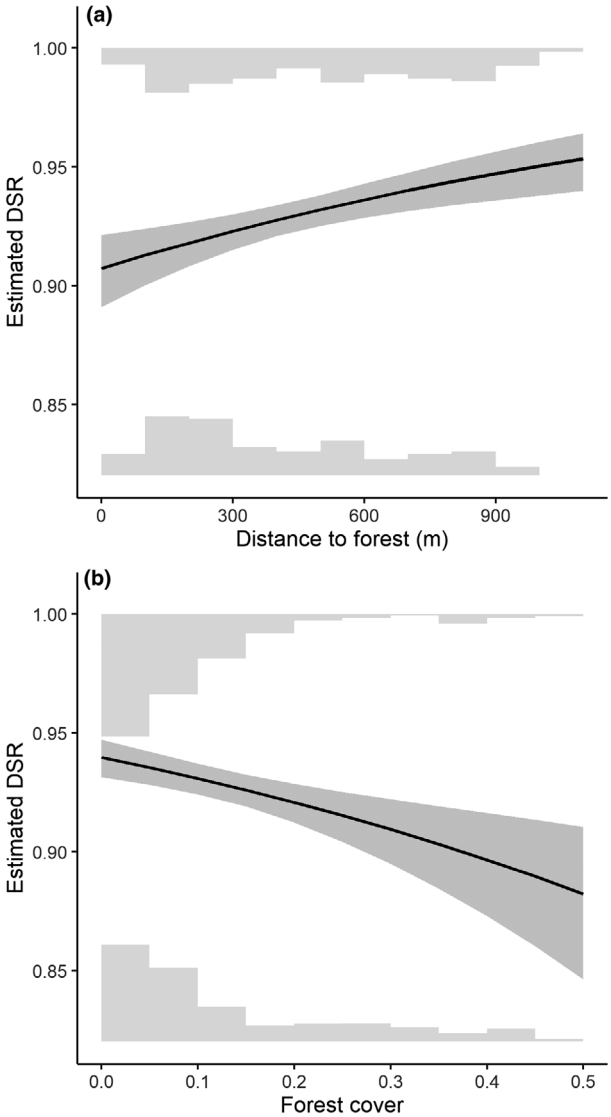

Figure 3 Estimated DSR in relation to distance to forest (a) and to proportion of forest cover within a 1-km buffer around each nest (b). The black line shows the predicted DSR values and grey area represents 95% confidence intervals. Histogram shows the distribution of the data, with hatched nests depicted at the top and depredated nests at the bottom of the plot.

believe that the proportion of nests with an unknown predator would have a different distribution than the depredated nests where the predator species was identified. Survival in nests with and without cameras did not differ (DSR with camera = 0.918 (SE = 0.010, 95% CI = 0.897–0.935), DSR without camera = 0.930 (SE = 0.003, 95% CI = 0.923–0.937).

Kruskal–Wallis test shows strong evidence that nests depredated by different predators differ in their distance to forest (X^2 = 18.04, df = 5, *P*-value = 0.003) and forest cover (X^2 = 22.77, df = 5, *P*-value <0.001). Predation by the most important predator, the red fox, occurred on average at 217 m (95% CI = 161–273 m) from the forest edge (Fig. 4). Although with very limited sample sizes, our results show that the northern raven also depredated nests close to the forest (mean distance = 181 m (95% CI = 119–241 m) while the European badger and golden jackal depredated nests further from the forest edge, at 606 (416–795 m) and 448 m (218–678 m), respectively (Fig. 4).

The only predator species with enough observations for reliable analysis of their association with forest edge was the red fox, recorded in 42 camera traps in use while too few camera traps had recordings of other species (golden jackal recorded in 12 cameras, northern raven in 15 and European badger in three). Recordings from camera traps can be grouped into two categories, from active (not predated, hatched nor abandoned) and inactive (between hatching/nest loss and the removal of camera) nests. Wilcoxon matched pairs test results show that there were more observations of the red fox during the active period of the nest (W = 5025, *P*-value = 0.004, mean active =0.239 (617 camera days), mean inactive = 0.118 (233 camera days)). This indicates that the red fox was drawn to nests that contained eggs and/or incubating adults. Only observations recorded during the time when the nest was active were used to assess the red fox detection rate in association with the forest edge. Based on the recordings of the camera traps, the red fox detection rate was higher closer to the forest edge (Fig. 5), but no similar relationship was found with the proportion of forest cover (Table 5).

Discussion

We studied nest survival of several wader species in a wet grassland habitat fragmented by forest. We demonstrate the edge effect on nest survival at edge and landscape level. At edge level, nest survival increases with the distance to the forest. As a landscape level effect, we show that the percentage of forest cover within a 1-km buffer around each nest

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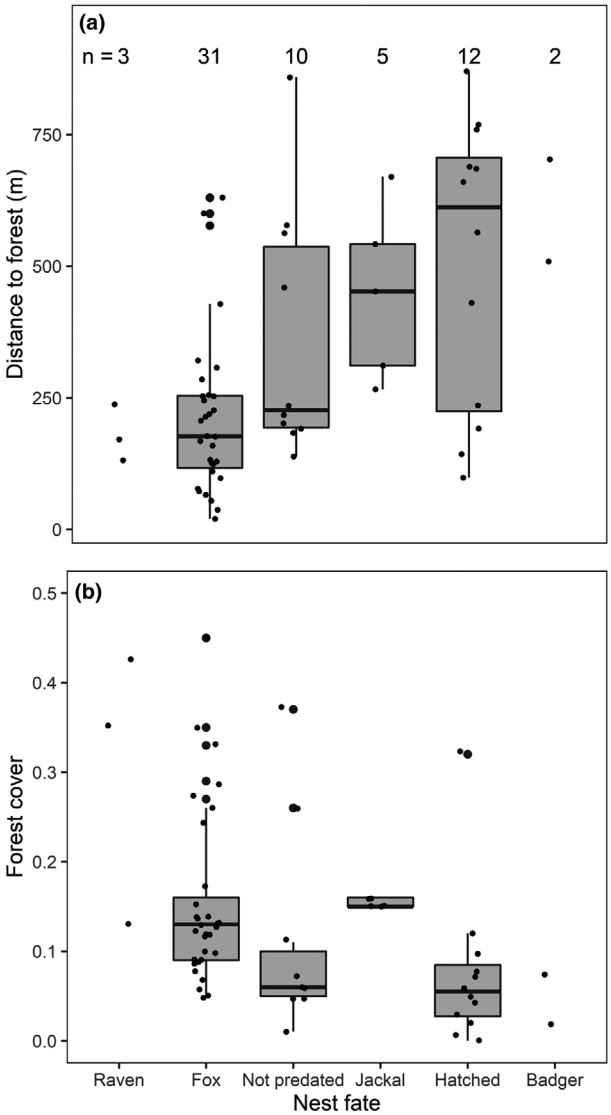

Figure 4 Nests predated by different species, hatched nests and nests that were not predated while the camera was following the nest (cameras were removed from sites when cattle was released) in relation to their distance to forest (a) and forest cover (b). Grey boxes indicate the interquartile range, horizontal line within the box denotes the mean value, whiskers show the minimum or maximum value of the data that is 1.5 times the interquartile range over the 75th percentile while large circles indicate outliers. Small black dots present data points. For species with less than five observations, only data points are presented.

negatively affects nest survival. The higher detection rate of the most common predator, the red fox, closer to the forest edge and the locations of the nests depredated by this species in the proximity of the forest suggest that the red fox is causing the edge effect in this study system. Our results also support the general findings that mammalian mesopredators are causing the majority of wader nest losses (MacDonald & Bolton, 2008a; McMahon *et al.*, 2020).

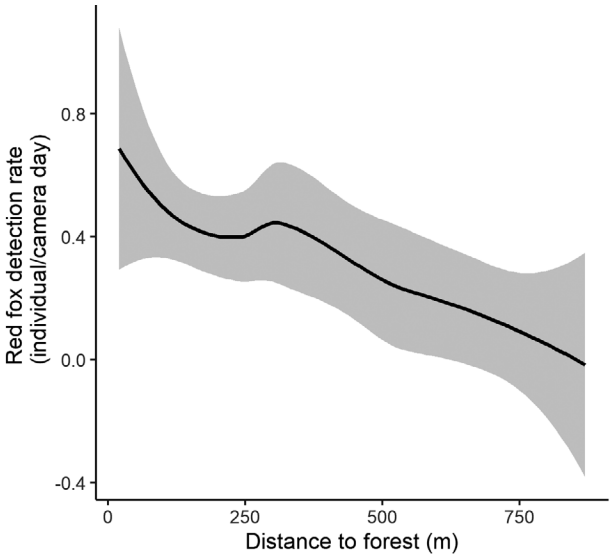

Figure 5 Locally weighted regression line showing the relationship between the distance to forest and the detection rate of red fox (expressed in unit individuals per camera day), grey area represents 95% confidence intervals.

The topic of edge effects has been widely covered in research, however, most of the studies have concentrated on explaining the edge-related changes in nest survival in forested landscapes. We demonstrate the edge effect on nest survival of ground-breeding birds in an open habitat surrounded by forest. Our results are in accordance with the findings of Ellis *et al.* (2020), Johnson & Stanley (1990) and Møller (1988), while Avery *et al.* (1989), Berg (1992), Berg *et al.* (1992b), Reino *et al.* (2010) and Vickery *et al.* (1992) have come to other conclusions. These contrasting results imply clearly that the effect of a forest edge in open landscapes is not straightforward, as also suggested by Donovan *et al.* (1997). Such variation may derive from the nesting behaviour of the bird species studied, the predator community (Lahti, 2001; Krüger, 2018), habitat type (Andrén, 1995; Chalfoun *et al.*, 2002), edge type (Pasitschniak-Arts *et al.*, 1998; Schneider *et al.*, 2012), woodland age (Hancock *et al.*, 2020) and distance from the edge considered (Paton, 1994; Chiavacci *et al.*, 2018; Ellis *et al.*, 2020). Moreover, it has been widely demonstrated that most wader species avoid breeding near forest edge, whereas the causal mechanism behind this is still not fully known (e.g. Stroud *et al.*, 1990; Amar *et al.*, 2011; Wilson *et al.*, 2014). Similar edge avoidance by breeding waders has been demonstrated in the Estonian coastal grassland habitat in an earlier study, while the cause was not investigated (Kaasiku *et al.*, 2019). Our study areas partially overlap (sharing seven out of 10 study sites) with the study where such behaviour was demonstrated, supporting the theory that the reason behind edge avoidance may be the experienced higher predation risk in the proximity of the forest edge.

While the decreasing distance to forest negatively affects wader nest success, our study reveals that the smaller

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Table 5 Generalized linear mixed model results testing the effect of distance to forest (Forest) and forest cover within a 1-km buffer around each nest (ForestCover), and Year on the detection rate of red fox (observations recorded by camera traps per day). Estimates for year are given in relation to 2019. Models significantly affecting the detection rate of red fox are marked in bold

Conditional model							Random effect		
	n		Estimate	SE	*z*	*P*		Variance	SD
Red fox detection rate	85	Intercept	−0.0017	0.061	−2.68	0.007	Site	0.402	0.634
		Forest	**−0.0004**	**0.001**	**−2.95**	**0.003**			
		ForestCover	0.0522	2.565	−0.02	0.984			
		Year	**0.8184**	**0.305**	**2.68**	**0.007**			
Zero inflation model		Intercept	−0.0017	0.683	−2.62	0.009			

patches of trees or bushes (<30 m wide) do not have an adverse effect on wader nest survival. Although some authors have combined these habitat features in their analyses (Ottvall *et al.,* 2005; Bertholdt *et al.,* 2017), we recommend future studies to approach smaller patches of trees and forest separately as they probably differ in their ecological function. It can be suspected that the patches of trees included in our analysis are too small to offer safe shelter for mammalian predators, nevertheless, they can still act as perches for avian predators. Our results contrast with the conclusions of Berg *et al.* (1992a) and Johansson (2001) who did describe lower nest survival closer to perches. In our study area, the only avian predator recorded depredating nests was the northern raven, whose hunting strategy in the study sites consists of scanning the area in flight rather than observing it from a perch (TK own observations). Hence, our result which shows that patches of trees do not increase predation, may only apply in areas where avian predators that tend to use perches are of little importance as nest predators. Also, if the conservation aim of the area is to support waders, removal of any woody vegetation is still recommended as these birds avoid nesting near trees (Stroud *et al.,* 1990; Hancock *et al.,* 2009).

In addition to the edge level effect, we also looked at nest survival in relation to surrounding landscape, investigating the effect of forest cover within a 1-km buffer around each nest. We conclude that the forest cover does have a negative effect on wader nest survival. Although landscape level effects on nest survival in open habitats have been little studied, similar result has been shown by Ellis *et al.* (2020), Valkama *et al.* (1999) and Douglas *et al.* (2014), who all have demonstrated lower nest success (or higher predation rates) in landscape that has more forest or lower proportion of breeding habitat. It has been suggested that forest cover can affect nest survival by supporting generalist predators that prefer mosaic areas and are less common in open landscapes (Reino *et al.,* 2010).

Besides exploring nest survival rates in relation to forest edge, we were also interested in understanding the possible mechanism behind it. In line with many other wader nest success studies, our results show the red fox as the main nest predator (e.g. Teunissen *et al.,* 2008; MacDonald & Bolton, 2008b; Bellebaum & Bock, 2009; Eglington *et al.,* 2009). The evidence presented in this study supports the interpretation that the edge effect is caused by the main nest predator foraging closer to forest edge: the detection rate of the red fox increases with decreasing distance to the forest and the nests predated by the red fox are on average close to the forest, at 217 m. Red fox density has been demonstrated to increase with proximity to forest and percentage of forest cover in British uplands (Douglas *et al.,* 2014; Hancock *et al.,* 2020). These findings possibly reflect the common behaviour of the red fox that hide in the forest but hunt in open areas (Lucherini *et al.,* 1995; Fiderer *et al.,* 2019), where their main food, rodents, is more plentiful (Soe *et al.,* 2017; Kana, 2019). However, it is important that such negative edge-associated effect is not carried over to other mammalian mesopredators as predator species possibly differ in their daily mobility and dependence of certain habitat features (Gehring & Swihart, 2003). Although with limited sample size, our results may suggest that both the European badger and golden jackal are not affected by the forest edge or that they operate at different spatial scales: the nests these species depredated were located further from the forest edge, averaging at 606 and 448 m, respectively. The possible association between these species and habitat composition should be further investigated.

Different mammalian predators possibly varying in their relation to forest edge may be the reason why relatively high levels of predation also occur in the nests furthest from the forest edge, at 1000 m (the maximum extent of our study areas). This is more than reported previously in other studies (400 m in Stroud *et al.* (1990) and < 150 m as reviewed by Laurance (2000)). More specifically, we show that while 7% of the nests closer to the forest hatch, the hatching rate is only 26% for the nests furthest from the forest. Likewise, 3% of nests hatch at highest forest cover, but even at the lowest forest cover, only 19% of the nests hatch. This result is a matter of serious concern and shows that habitat fragmentation may have more severe effects on the open landscape species than realized before. As the Estonian coastal grassland habitat mostly lies as a narrow strip between the sea and the higher wooded parts and rarely exceeds 1 km in width, the 26% hatching rate is unlikely to be exceeded anywhere. Moreover, we also show that no species is likely to escape this fate: while vegetation structure around the nest can affect nest survival (Laidlaw *et al.,* 2020), species with open (northern lapwing, common ringed plover) and

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concealed nests (common redshank, southern dunlin) with high enough sample size for species-specific estimations showed similarly low daily nest survival values in our study (Table 2). This may indicate that birds' behavioural adaptations such as specific microhabitat selection (open vs. concealed nests) or nest defence strategy (mobbing vs. quietly fleeing) may not affect the nest predation rates.

Our study design is limited as it does not include grassland vegetation characteristics as covariates that could affect nest survival. Previously, vegetation height and heterogeneity have been shown to affect rodent and predator densities (Klug *et al.,* 2009; Schneider *et al.,* 2012; Laidlaw *et al.,* 2019). Nonetheless, we hope the equal susceptiveness to nest predation by wader species with different requirements for vegetation height (Durant *et al*., 2008; Stroud *et al.,* 1990) alleviates the concern that vegetation would interfere with nest predation patterns found in this study.

Conservation implications

From the one hand, our results raise caution about the afforestation of open habitats in regions where nest predators benefit from forest cover. These plantations can negatively affect threatened ground-breeding birds in an area much wider than occupied by the woody vegetation itself. From the other hand, such result means that habitat conservation and restoration of open landscapes needs to aim for achieving wide intact areas. Nevertheless, our results also imply, similarly to Smart *et al*. (2013) and Douglas *et al*. (2014), that habitat restoration by removal of trees alone may not be sufficient to overcome the low wader nest survival, especially in areas where the possibilities to extend the width of the habitat are limited. Hence, it is important to consider the implementation of other predator control methods alongside habitat restoration, at least until habitat restoration is implemented. Common predator control practices include predator exclusion fences and targeted culling of predators near important conservation areas, which have been shown to have a varying but overall positive effect on nest survival (Smith *et al.,* 2010, 2011). However, removal of forest plantations should still remain a priority, as it has been shown that the effectiveness of predator control interacts with the amount of forest in the landscape (Douglas *et al.,* 2014).

Our results can be directly applied to improve the state of the breeding grounds of threatened wader species in Estonian coastal grasslands. Currently, these open habitat specialists have been pushed to survive in small pockets of the remaining habitat, that only 60 years ago spanned a considerably wider area. These remnant sites are now managed under an agri-environment scheme, with the aim to maintain suitable habitat for breeding waders. Although the agri-environment scheme applied has achieved positive results in vegetation management (Kaasiku *et al.,* 2021), increasing the suitable area for breeding waders, it cannot be truly effective if the sites are entirely affected by the edge effect. If no effort is made to achieve the former extent of the area or manage the low nest survival in other ways, these wader populations may not sustain for much longer. As similar issues threaten open habitat specialists elsewhere where predator community benefits from fragmented landscapes, removal of fragmenting elements can be assumed to have a positive effect on these species in different parts of the world (Wilson *et al.,* 2014; Ellis *et al.,* 2020). We call for large scale trials that remove woodlands from important wader breeding areas to test this assumption.

The findings of this study support the previous knowledge that the increase of forest edge and the expansion of generalist predators are two changes of anthropogenic origin which in combination can have a profoundly damaging effect on ground-breeding birds (Stephens *et al.,* 2004; McMahon *et al.,* 2020). Policies that support current tree plantations or forestry expansion need to be aware that fragmentation of open landscapes has serious detrimental consequences to open habitat specialists such as breeding waders. At the same time, supporting restoration efforts that decrease the extent of forest plantations may benefit breeding waders.

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Supporting information

Additional supporting information may be found online in the Supporting Information section at the end of the article.

Table S1. Beta estimates, standard error (SE), lower confidence limit (LCL), and upper confidence limit (UCL) for all nest survival models.

Figure S1. Nest timings depicted by site (rows) and year (columns), grey vertical bars denote active nests from laying to hatching or predation event.

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